Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Advanced Algorithms and Parallel Programming – II part –January 26th 2024

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and briefly explain (without an explanation, the answer will be considered invalid)

A. Is it possible to eliminate a flow dependency by modifying the program and ensuring that the program remains equivalent? Is it the same with output dependency? Please show an example of what can be done. (2)

B. Under what conditions can the following code be parallelized? If these conditions are met, is there a way to write the code for better parallelization? (2)

```
for (i=1; i<100; i++)
        for (j=0; j<100; j++)
            c[j][i] += a[i]*b[j];
```

Exercise n. 2

Answer the following questions about parallel patterns and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe the parallel gather pattern: assumptions, complexity, and example. (2)

B. Please describe how the "Merge collision resolution" works. Is it related to the parallel gather pattern? (2)

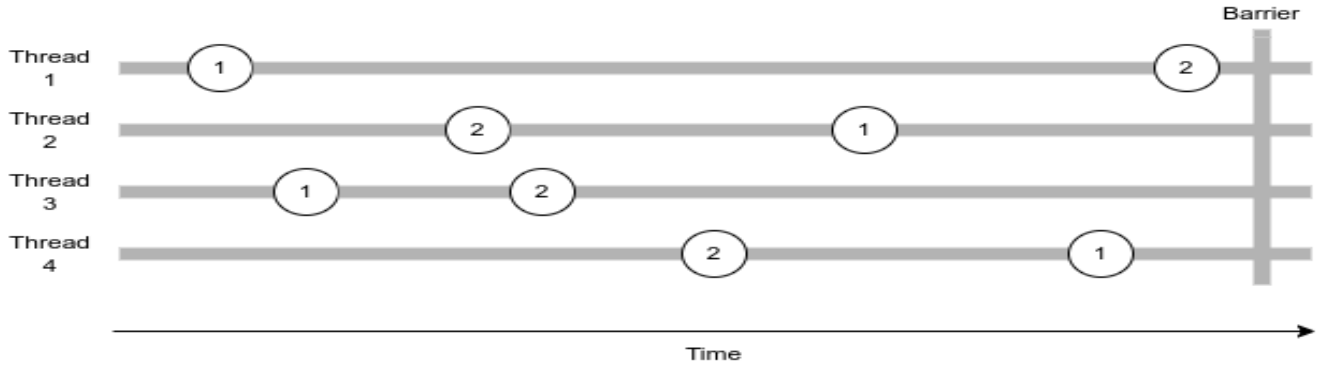
Exercise n. 3

Answer the following questions about shared memory parallel programming and briefly explain (without an explanation, the answer will be considered invalid)

A. What is the difference between joinable and detached threads in Pthreads? (1)

B. What is the difference between joining and waiting on a barrier in Pthreads? (1)

C. Describe how each thread in the following diagram behaves at runtime as it encounters points marked with 1, representing `pragma omp cancel`, and with 2, representing `pragma omp cancellation point`. (2)



Exercise n. 4

Answer the following questions about MPI and briefly explain (without an explanation, the answer will be considered invalid)

A. What are the approaches that the receiver in point-to-point communications can use to allocate a suitable receiving buffer, i.e. allocate a suitable number of elements? (1) **This question can be skipped if you passed the challenge.**

B. Why are there multiple MPI implementations? (1)

C. What are the main differences between collective and point-to-point functions when we focus on data exchange functions? (2)